

# NERIC JOEL

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## Profile

Computer Science master's student and AI/ML engineer skilled in Python, PyTorch, and agentic AI systems, including tool-calling, RAG pipelines, and prompt design. Experienced deploying ML models and agent-based workflows, and mentoring 120+ students across AI and programming, turning real-world data into high-impact tools.

## Education

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| <b>Arizona State University</b> , Tempe, AZ                                                               | <i>Aug 2025 – May 2027</i>     |
| Master of Science in Computer Science                                                                     | Cumulative GPA: 3.66/4.0       |
| Relevant Coursework: Machine Learning, Deep Learning, Advanced Algorithms, Distributed Systems, Databases |                                |
| <b>Amrita Vishwa Vidyapeetham</b> , Coimbatore, India                                                     | <i>Sep 2021 – May 2025</i>     |
| Bachelor of Technology in Electrical and Electronics Engineering                                          | First Class, Merit Scholarship |

## Professional Experience

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| <b>W. P. Carey School of Business, Arizona State University</b> , Tempe, AZ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>May 2026 – July 2026</i> |
| <i>Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                             |
| <ul style="list-style-type: none"><li>Debugged system prompts, tool-calling routines, and REST API integrations for AI agents on the n8n automation platform, supporting 106 graduate students and resolving most issues within the same session.</li><li>Explained agentic AI concepts (chat completions, function calling, prompt design) to students from non-technical business backgrounds, both one-on-one and in small groups.</li><li>Authored reference solutions and debugging guides that cut repeat questions and standardized how the cohort validated their agents.</li></ul> |                             |
| <b>PV Systems Lab, Amrita University</b> , Coimbatore, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>Jan 2024 – May 2025</i>  |
| <i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                             |
| <ul style="list-style-type: none"><li>Built and benchmarked predictive models in Python and SQL (RMSE/MAE against experimental data), raising degradation-forecast accuracy 18% over the baseline.</li><li>Cleaned and analyzed 100,000+ records of bench and field telemetry, then built automated ETL pipelines that cut manual data prep 40%.</li><li>Developed Tableau visualizations and regression models to surface reliability and field-failure patterns, improving operational efficiency 15%.</li></ul>                                                                          |                             |

## Leadership & Extracurriculars

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| <b>AI Club, Arizona State University</b> , Tempe, AZ                                                                                                                                                    | <i>Aug 2025 – Present</i>  |
| <i>Member</i>                                                                                                                                                                                           |                            |
| <ul style="list-style-type: none"><li>Collaborate on applied AI/ML and computer-vision research initiatives, and mentored 15+ students in Python, SQL, and machine-learning fundamentals.</li></ul>     |                            |
| <b>IEEE Student Branch</b> , Coimbatore, India                                                                                                                                                          | <i>Sep 2022 – May 2025</i> |
| <i>Volunteer &amp; Co-Author</i>                                                                                                                                                                        |                            |
| <ul style="list-style-type: none"><li>Organized technical workshops and stakeholder-facing sessions; co-authored two IEEE papers on PV parameter estimation and low-power CMOS design (2025).</li></ul> |                            |

## Tech Projects

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| <b>Agentic AI Case Management System</b>                                                                                                                                                                                                                                                                                                                                | <i>Python, Tool-Calling, WiCS × Opportunity Hack</i> |
| <ul style="list-style-type: none"><li>Built a case-management tool driven by agentic AI workflows, using tool-calling and structured outputs to automate intake and routing for a nonprofit during a weekend hackathon.</li><li>Wrote and iterated on system prompts and API integrations so the agent stayed reliable and on task across varied user inputs.</li></ul> |                                                      |
| <b>Building Footprint Segmentation</b>                                                                                                                                                                                                                                                                                                                                  | <i>PyTorch, U-Net + ResNet50, Computer Vision</i>    |
| <ul style="list-style-type: none"><li>Built a deep-learning pipeline (U-Net with a ResNet50 encoder) in PyTorch to extract building footprints from aerial imagery on the Massachusetts Buildings Dataset.</li><li>Tuned architecture and hyperparameters against IoU and precision-recall, and profiled the model to cut inference latency 30%.</li></ul>              |                                                      |
| <b>AI-Driven Solar PV Emulator</b>                                                                                                                                                                                                                                                                                                                                      | <i>Python, Neural Networks, Predictive Modeling</i>  |
| <ul style="list-style-type: none"><li>Processed and transformed 120,000+ time-series records for supervised learning and neural-network training.</li><li>Trained neural-network and regression models with numerical optimization, reducing prediction error 18% versus the baseline (RMSE, MAE).</li></ul>                                                            |                                                      |

## Skills & Tools

**Tools:** Python, SQL, C++, PyTorch, TensorFlow, scikit-learn, Hugging Face, n8n, FastAPI, PostgreSQL, Git, Tableau, Claude/OpenAI API

**Skills:** Machine Learning, Deep Learning, Agentic AI & Tool-Calling, RAG Pipelines, System Prompt Design, Prompt Engineering, Model Evaluation (RMSE/MAE, IoU), ETL / Data Pipelines, Statistical Analysis